| Question | | | Answer | Marks | AO | Guidance |
|---|---|---|---|---|---|---|
| 17 | | | $\frac{dy}{dx} = 2x$ | **M1** | **1.1** | Accept $2t$ |
| | | | Gradient of normal is $-\frac{1}{2t}$ | **M1** | **1.1** | For use of negative reciprocal oe |
| | | | $y - t^2 = -\frac{1}{2t}(x - t)$<br>So $y = -\frac{x}{2t} + t^2 + \frac{1}{2}$ | **A1** | **2.1** | For convincingly reaching given result<br>Any error seen is **A0**. |
| | | | **Alternative method for final mark** | | | |
| | | | For the given line when $x = t$,<br>$y = -\frac{1}{2} + t^2 + \frac{1}{2} = t^2$ and the gradient is $-\frac{1}{2t}$ | **A1** | | |
| | | | | **[3]** | | |

**Q17.** The first M1 is for differentiating to get either 2x or 2t.
The second M1 is for using the negative reciprocal to get -1/their gradient.
The A1 is for substituting into $(y - y_1) = m(x - x_1)$ or $y = mx + c$ to get the required result or equivalent unsimplified. It must be convincing as the answer is given. Any error seen is A0.